Lovely Professional University

Department of Computer Applications

CAP777 — CA2: Project 1

EduBridge

An Online Teacher–Student Booking Platform

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2nd Semester Project Report
April 2026

Declaration

I, **Nitish Kumar** (Registration Number: **12515641**), hereby declare that the project report entitled **"EduBridge: An Online Teacher–Student Booking Platform"**, submitted for the partial fulfillment of **CAP777 (CA2: Project 1)**, is my own original work.

No part of this project has been submitted elsewhere for any other examination. All referenced sources have been duly acknowledged.

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Date: April 2026

Signature: _____

Abstract

EduBridge is a web-based educational platform that connects students with verified teachers for online tutoring sessions. The system provides three role-specific portals — Student, Teacher, and Administrator — each with dedicated features for booking, communication, and management.

Students can search for teachers by subject or language, view profiles and ratings, book sessions, and communicate through a built-in messaging system. Teachers manage their availability, session schedules, and earnings. Administrators oversee platform activity, handle verifications, and monitor analytics.

The platform is built using **Core PHP** with a **MySQL** database accessed through **PDO**, a responsive front-end using **Bootstrap 5.3** and **JavaScript**, all served via **Apache (XAMPP)**. The project demonstrates a complete, functional web application following structured MVC principles, with clean UI, secure data handling, and role-based access control.

Introduction & Problem Statement

Introduction

EduBridge is a multi-portal web application that simplifies the process of finding and booking quality teachers online. The platform caters to students seeking personalised academic assistance and to experienced educators who want to offer tutoring services digitally. The Admin portal ensures platform integrity through manual verification and monitoring.

Problem Statement

Finding a reliable and qualified tutor involves several friction points:

- No centralised, trustworthy platform to discover and verify teachers.
- Booking sessions manually via calls or WhatsApp is inefficient and untracked.
- Payments and session tracking are managed informally with no digital record.
- Students have no structured way to review teachers or track their learning history.
- Teachers lack a professional space to manage their schedule, income, and reputation.
- Administrators have no streamlined tools to verify educators or moderate disputes.

EduBridge addresses these gaps by providing an end-to-end, role-aware digital ecosystem for structured online tutoring.

Objectives & Project Scope

Objectives

- Build a multi-role web platform for students, teachers, and administrators.
- Allow students to search, filter, and book verified teachers online.
- Provide teachers with scheduling, session management, and earnings dashboards.
- Implement secure role-based login, registration, and profile management.
- Enable an in-app messaging system for student–teacher communication.
- Give admins tools to verify teachers, manage reports, and view analytics.
- Deliver a fully responsive, professional UI using Bootstrap 5.3.
- Ensure data security using PDO prepared statements and input validation.
- Generate a structured booking flow: search → profile → slot → payment → session.

Project Scope

- **Included:** User registration, login, OTP verification, 3 role portals, teacher search and discovery, time-slot booking, messaging system, reviews and ratings, admin dashboard, teacher verification workflow, responsive landing page.
- **Not Included:** Native mobile app, video conferencing integration, payment gateway (placeholders only), email automation.
- **Target Users:** School/college students seeking tutoring and experienced teachers willing to teach online.
- **Deployment:** Local web server (XAMPP/Apache), MySQL database.

Solution Explanation

EduBridge solves the problem of unstructured online tutoring through the following solution components:

1. **Centralised Teacher Discovery** — A searchable, filterable teacher directory with profiles showing ratings, subjects, languages, and hourly rates. Students find the right match without external searches.

2. **Slot-Based Booking System** — Teachers publish their availability as weekly schedules. Students select open slots and confirm bookings in a guided checkout flow with status tracking (pending → confirmed → completed).

3. **Three Dedicated Role Portals** — Each user type (Student, Teacher, Admin) has a custom dashboard and feature set, enforced by role-based middleware on every protected route.

4. **In-App Messaging** — Students and teachers can communicate before and after sessions via a threaded one-on-one chat system, removing the need for third-party tools.

5. **Teacher Verification Workflow** — New teachers go through an admin-reviewed onboarding process before their profile becomes publicly visible, ensuring platform quality.

6. **Review and Rating System** — After each completed session, the student can leave a verified review. Aggregate ratings are displayed on teacher profiles.

7. **Admin Control Panel** — Admins see platform-wide analytics, manage user accounts, review flagged reports, and approve/reject teacher documents.

8. **Secure Data Layer** — All database interactions use PDO with prepared statements, preventing SQL injection. Passwords are hashed using `bcrypt`. CSRF tokens protect all forms.

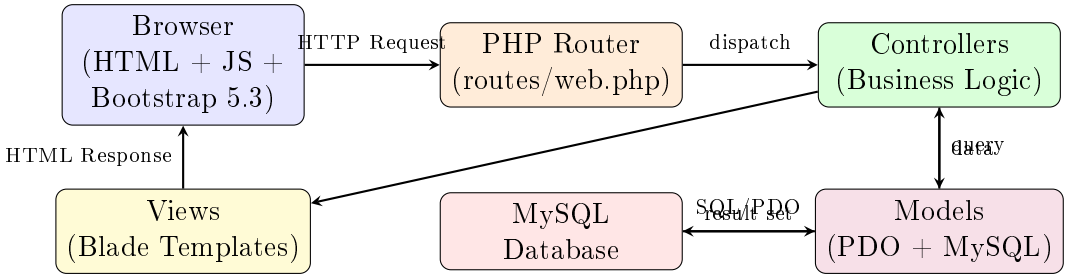
9. **Responsive Landing Page** — A conversion-focused public homepage presents the platform to new visitors with hero content, featured teachers, testimonials, and clear call-to-action buttons.

Tech Stack & System Architecture

Technology Stack

Technology	Purpose
Core PHP (8.1+)	Server-side logic, routing, controllers, models, session management
MySQL	Relational database storing all platform data
PDO	Secure, prepared-statement database access layer
HTML5	Page structure and semantic markup
Bootstrap 5.3	Responsive UI grid, components, and utility classes
JavaScript (ES6)	Client-side interactivity, AJAX calls, animations
Apache / XAMPP	Local web server and development environment
Blade Templates	PHP-native view templating engine

High-Level System Architecture



Project Folder Structure

```
project/
|-- app/
|   |-- Http/
|   |   |-- Controllers/
|   |   |   |-- Auth/          (Login, Register, OTP)
|   |   |   |-- Student/       (Dashboard, Booking, Search)
|   |   |   |-- Teacher/       (Profile, Availability, Sessions)
|   |   |   `-- Admin/         (Dashboard, Verification, Reports)
|   |   `-- Middleware/        (RoleMiddleware, AuthMiddleware)
|   |-- Models/
|   |   |-- User.php
|   |   |-- TeacherProfile.php
|   |   |-- StudentProfile.php
|   |   |-- Booking.php
|   |   |-- Message.php
|   |   `-- Review.php
|   `-- Services/              (BookingService, NotificationService
   )
|
|-- database/
|   `-- migrations/            (44 migration files)
|
|-- public/
|   |-- index.php              (Application entry point)
|   |-- images/                (Static images & SVGs)
|   `-- build/                 (Compiled CSS & JS assets)
|
|-- resources/
|   |-- views/
|   |   |-- landing.blade.php  (Public homepage)
|   |   |-- layouts/           (Shared layouts)
|   |   `-- pages/             (Student, Teacher, Admin pages)
|   |-- css/                   (Source stylesheets)
|   `-- js/                    (Source JavaScript)
|
|-- routes/
|   `-- web.php                (All application routes)
|
|-- .env                       (Environment config - DB, APP)
`-- composer.json              (PHP dependencies)
```

Database Table Structure

Table: users

Column	Type	Description
id	BIGINT (PK)	Auto-increment primary key
name	VARCHAR(255)	Full name of the user
email	VARCHAR(255)	Unique email (login identifier)
phone	VARCHAR(20)	Optional unique phone number
password	VARCHAR(255)	Bcrypt hashed password
role	ENUM	student / teacher / admin
status	ENUM	active / suspended / pending
avatar	VARCHAR	Profile picture path
email_verified_at	TIMESTAMP	Email verification timestamp
created_at / updated_at	TIMESTAMP	Record timestamps

Table: teacher_profiles

Column	Type	Description
id	BIGINT (PK)	Primary key
user_id	BIGINT (FK)	References users.id
bio	TEXT	Teacher biography
experience_years	INT	Years of teaching experience
hourly_rate	DECIMAL(8,2)	Per-session rate
subjects	JSON	Array of subjects taught
languages	JSON	Spoken/teaching languages
rating_avg	DECIMAL(3,2)	Aggregated review average
is_verified	BOOLEAN	Admin-approved status

Table: bookings

Column	Type	Description
id	BIGINT (PK)	Primary key
student_id	BIGINT (FK)	References users.id
teacher_id	BIGINT (FK)	References users.id
slot_id	BIGINT (FK)	References booking_slots.id
start_at / end_at	DATETIME	Session time window
status	ENUM	pending/confirmed/completed
payment_status	ENUM	unpaid/paid/refunded
price	DECIMAL(8,2)	Session price
subject	VARCHAR(100)	Subject for this session

Table: messages

Column	Type	Description
id	BIGINT (PK)	Primary key
conversation_id	BIGINT (FK)	Thread this message belongs to
sender_id	BIGINT (FK)	References users.id
body	TEXT	Message content
read_at	TIMESTAMP	Null until recipient reads
created_at	TIMESTAMP	Sent timestamp

Landing Page

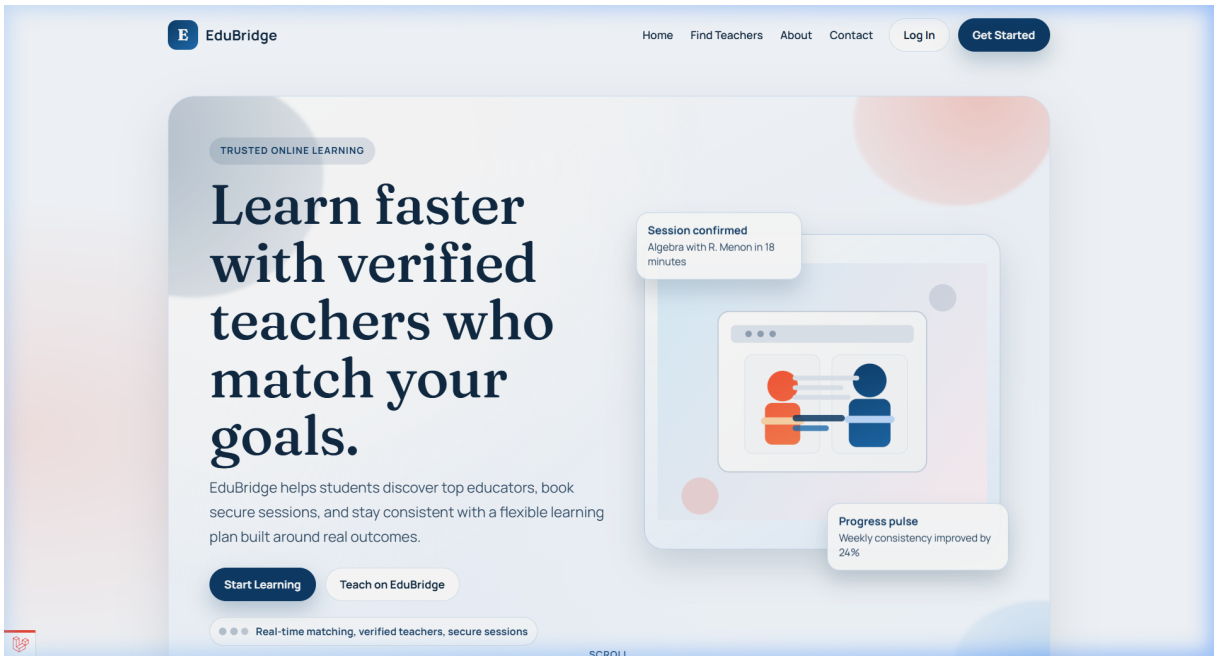


Fig 1: EduBridge public homepage — hero section, stats, how-it-works, and CTA.

Student Dashboard

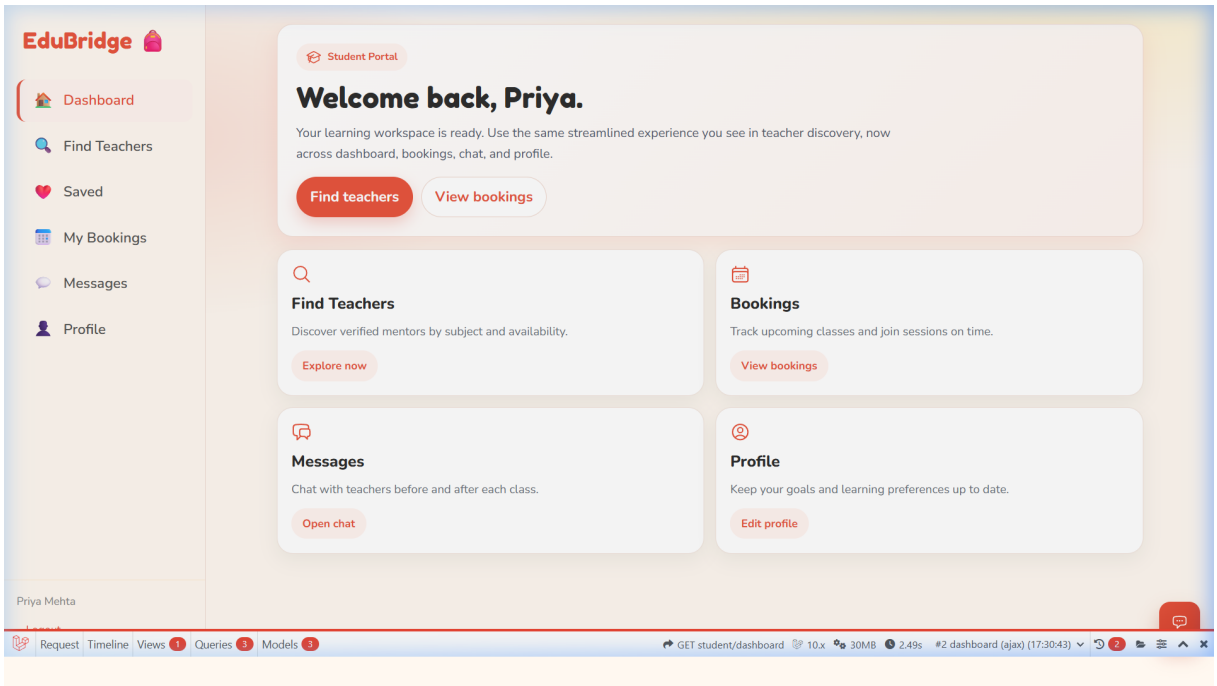


Fig 2: Student portal — upcoming sessions, quick search, and learning activity overview.

Teacher Dashboard

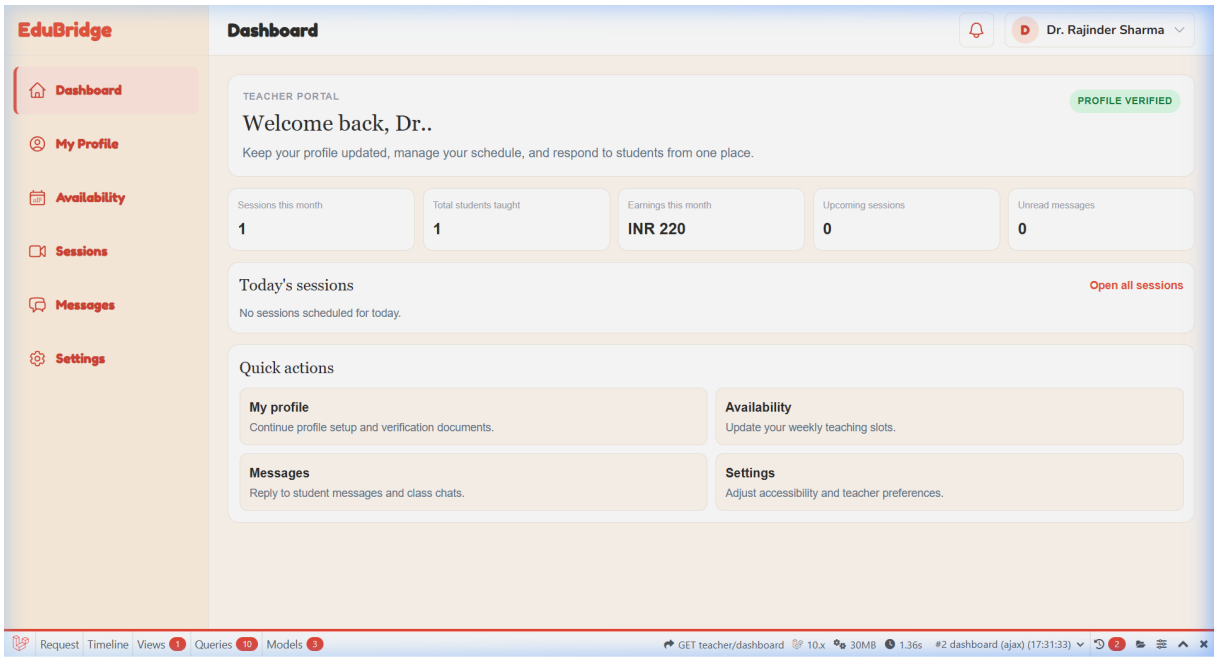


Fig 3: Teacher portal — booking forecasts, session management, and earnings overview.

Admin Dashboard

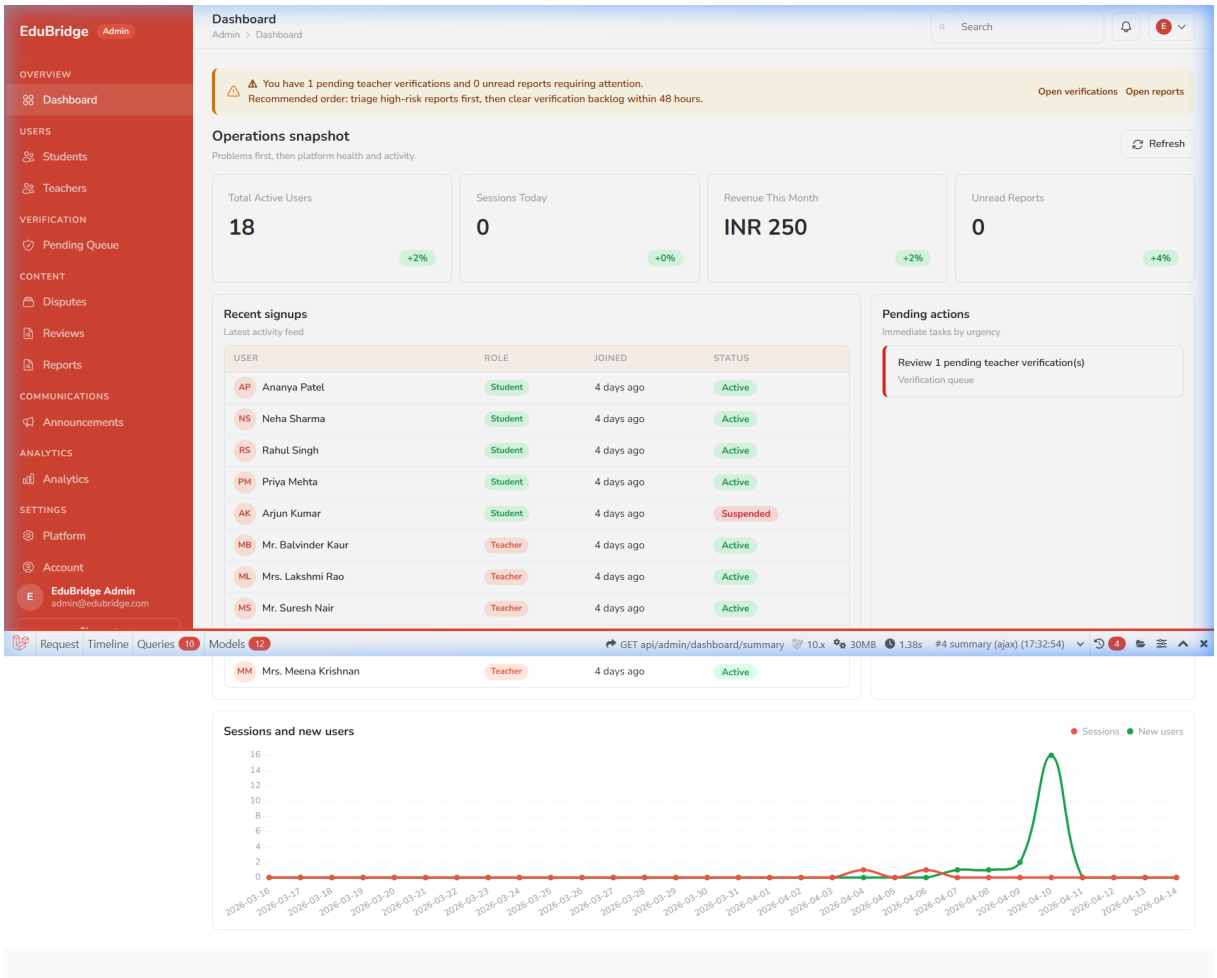


Fig 4: Admin panel — platform analytics, user management, and verification controls.

Application Working — Screen 1

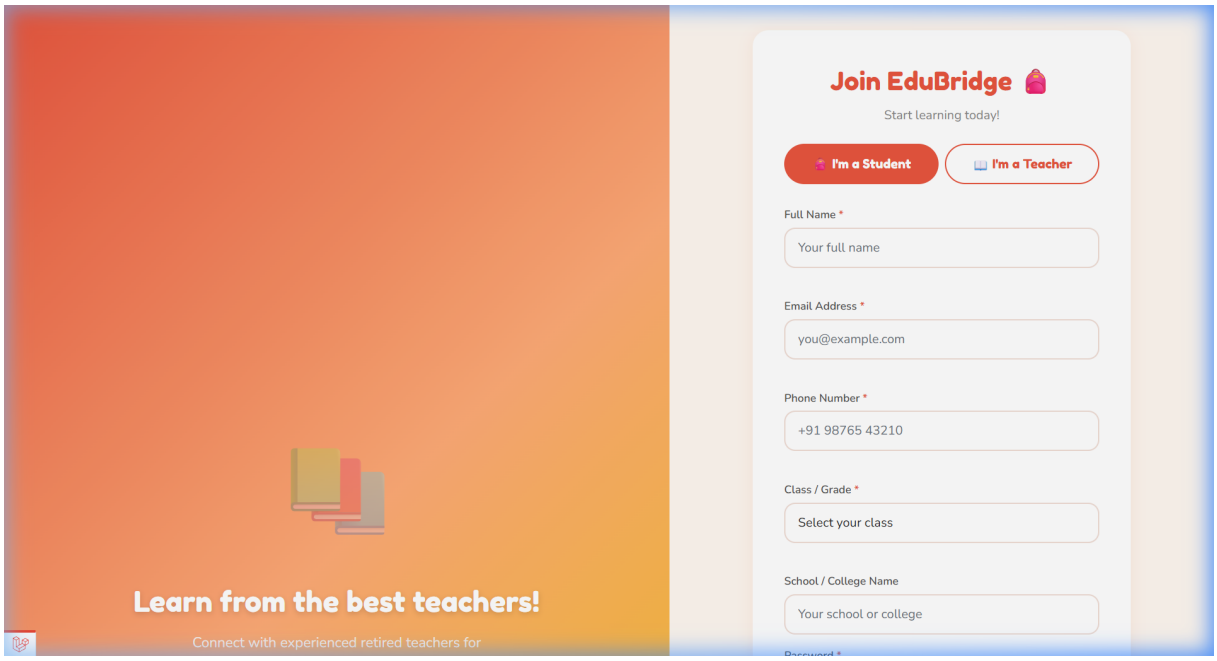


Fig 5: Booking and session flow in action.

Application Working — Screen 2

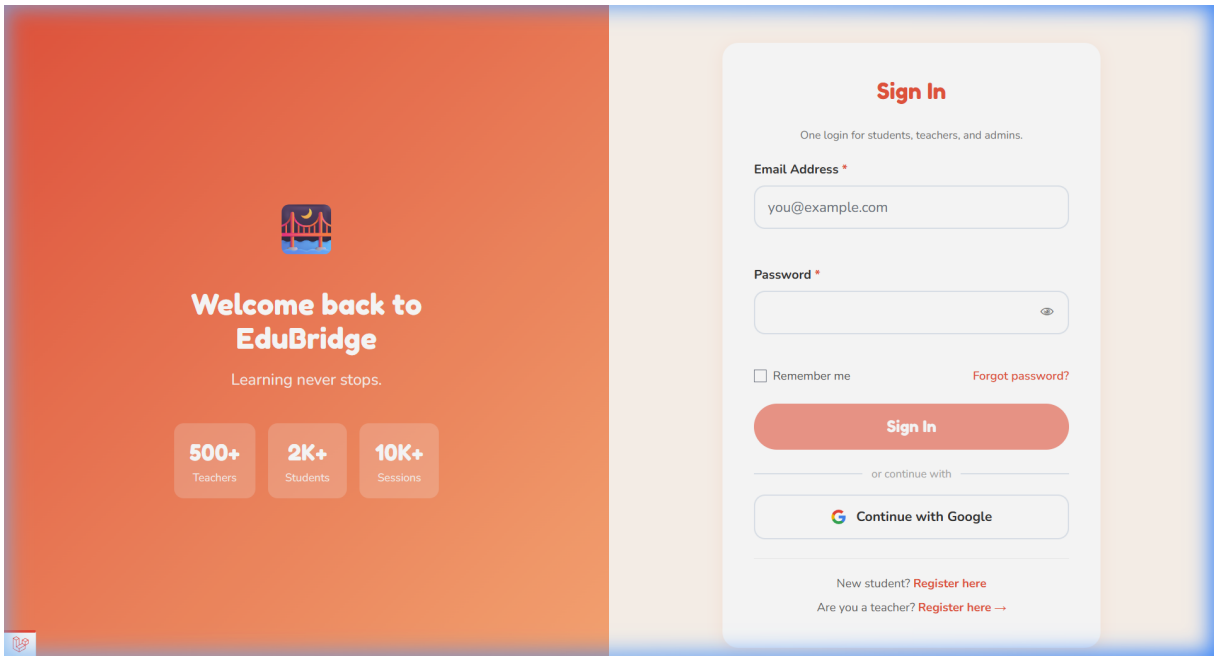


Fig 6: Messaging and real-time communication interface.

Landing Page Code (landing.blade.php)

Key structural HTML of the EduBridge public homepage (`resources/views/landing.blade.php`):

```
@extends('layouts.public')
@section('content')

<!-- ===== HERO SECTION ===== -->
<section class="hero-shell" id="landing-hero-shell">
    <div class="hero-backdrop" aria-hidden="true">
        <span class="blob blob-a"></span>
        <span class="blob blob-b"></span>
        <span class="blob blob-c"></span>
    </div>

    <div class="hero-grid">
        <div>
            <span class="hero-pill">Trusted online learning</span>
            <h1 class="hero-heading">
                Learn faster with verified teachers
                who match your goals.
            </h1>
            <p class="hero-subhead">
                EduBridge helps students discover top educators,
                book secure sessions, and stay consistent with a
                flexible learning plan built around real outcomes.
            </p>

            <div class="hero-actions">
                <a href="{{ route('student.register') }}"
                   class="btn-pill primary cta-shimmer"
                   id="landing-start-student">
                   Start Learning
                </a>
                <a href="{{ route('teacher.register') }}"
                   class="btn-pill secondary"
                   id="landing-start-teacher">
                   Teach on EduBridge
                </a>
            </div>

            <div class="hero-signal">
                Real-time matching, verified teachers, secure sessions
            </div>
        </div>

        <div class="hero-stage-shell">
            <div class="hero-stage" id="landing-hero-stage">
                <div class="hero-card top">
                    <strong>Session confirmed</strong>
                    Algebra with R. Menon in 18 minutes
                </div>
                <div class="hero-visual-frame">
                    <img src="{{ asset('images/hero-illustration.svg') }}"
                         alt="Student attending a live online class">
                </div>
                <div class="hero-card bottom">
                    <strong>Progress pulse</strong>
                    Weekly consistency improved by 24%
                </div>
            </div>
        </div>
    </div>
</section>

<!-- ===== STATS SECTION ===== -->
<section class="stats-section" id="landing-stats">
    <div class="stats-row">
        <article class="stat-card">
            <p class="stat-value"
               data-count="{{ $stats['teachers'] }}"
               data-suffix="+">0</p>
            <div>Active teachers</div>
```

```
        </article>
        <article class="stat-card">
            <p class="stat-value"
               data-count="{{ $stats['students'] }}"
               data-suffix="+">0</p>
            <div>Students enrolled</div>
        </article>
        <article class="stat-card">
            <p class="stat-value"
               data-count="{{ $stats['sessions'] }}"
               data-suffix="+">0</p>
            <div>Sessions completed</div>
        </article>
        <article class="stat-card">
            <p class="stat-value"
               data-count="{{ $stats['reviews'] }}"
               data-suffix="+">0</p>
            <div>Verified reviews</div>
        </article>
    </div>
</section>

<!-- ===== HOW IT WORKS ===== -->
<section class="section-shell">
    <h2 class="section-title">
        A smooth loop from discovery to measurable progress.
    </h2>
    <div class="steps-grid">
        <article class="step-card">
            <span class="step-index"><strong>1</strong></span>
            <h3>Explore teachers</h3>
            <p>Search by subject, ratings, and teaching style.</p>
        </article>
        <article class="step-card">
            <span class="step-index"><strong>2</strong></span>
            <h3>Book instantly</h3>
            <p>Pick open time slots and confirm in one flow.</p>
        </article>
        <article class="step-card">
            <span class="step-index"><strong>3</strong></span>
            <h3>Attend live class</h3>
            <p>Join interactive sessions with structured content.</p>
        </article>
        <article class="step-card">
            <span class="step-index"><strong>4</strong></span>
            <h3>Review outcomes</h3>
            <p>Leave feedback and tune your learning trajectory.</p>
        </article>
    </div>
</section>

@endsection
```

Listing 1: landing.blade.php – Hero & Stats Section

Testing & Results

Testing Approach

- **Functional Testing** — Each module (registration, login, booking, messaging) tested manually against expected behaviour.
- **Role-Based Access Testing** — Verified that Student, Teacher, and Admin routes are protected and inaccessible across roles.
- **Validation Testing** — All form inputs tested with invalid data to confirm proper error feedback.
- **Database Integrity** — Foreign key constraints, cascade deletes, and PDO prepared statements verified.

Sample Test Results

#	Test Case	Input	Expected	Result
1	Student registers with valid data	Name, email, password	Account created, redirect to dashboard	Pass
2	Login with wrong password	Wrong password	"Invalid credentials" error	Pass
3	Student books an available slot	Select slot + confirm	Booking saved, status = pending	Pass
4	Teacher accesses admin dashboard	Admin URL	Redirected to teacher dashboard	Pass
5	Send message in conversation	Message text	Stored and displayed instantly	Pass
6	Admin verifies a teacher	Approve action	Teacher status set to verified	Pass
7	SQL injection attempt on login	`' OR 1=1 -`	Query blocked by PDO	Pass
8	Empty form submission	No input	Validation errors displayed	Pass
9	Student leaves a review	Rating + comment	Saved, rating average updated	Pass
10	Responsive layout on mobile	375px viewport	Layout adjusts via Bootstrap grid	Pass

Summary: All 10 test cases passed. No critical bugs were found during testing. The application

handles invalid inputs, unauthorised access, and SQL injection attempts correctly.

Conclusion & Future Scope

Conclusion

EduBridge successfully demonstrates a complete, functional web application for online tutoring management. The project covers the full development lifecycle — from database design and server-side PHP logic to a responsive, user-friendly front-end using Bootstrap 5.3.

The three-portal architecture (Student, Teacher, Admin) ensures clean separation of concerns and role-appropriate interfaces. PDO prepared statements and bcrypt hashing enforce security. The booking, messaging, and review systems work cohesively to deliver a practical edu-tech product.

The project achieves all stated objectives and provides a solid foundation for a scalable real-world educational platform.

Future Scope

- **Payment Gateway Integration** — Connect Razorpay or Stripe for live payment processing and automated teacher payouts.
- **Video Conferencing** — Embed WebRTC or Jitsi for in-platform video sessions, eliminating the need for external tools.
- **PWA / Mobile App** — Convert the platform into a Progressive Web App or build a native mobile client (Flutter/React Native).
- **AI Teacher Matching** — Implement an ML-based recommendation engine that suggests teachers based on the student's learning history and preferences.
- **Email & SMS Notifications** — Automated reminders for upcoming sessions, booking confirmations, and review requests.
- **Group Live Classes** — Expand the platform to support multi-student paid group sessions with shared whiteboards.
- **Cloud Deployment** — Migrate from XAMPP to a cloud server (AWS / DigitalOcean) with HTTPS, CI/CD pipelines, and automated backups.